Essential Readings

HEALTH ECONOMICS AND FINANCING
ELECTIVE PAPER
M.A/M.Sc/ MBD/MSD, 2025-27 (E4.2/E4.3)
11th August, 2026

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Essential Readings

- Rexford E. Snterre and Stephen P. Neun. ***Health Economics: Theories, Insights and Industry Studies*** (614, San/Hea, 073226)

- Folland, Goodman, Stano & Danagoulian (2024), The Economics of Health and Health Care

- Drummond MF, Sculpher MJ, Torrance GW, O’Brien B, Stoddart GL eds. ***Methods for Economic Evaluation of Health care Programmes*****,** Third Edition, Oxford University Press, 2005.

- O'Donnell O, Doorslaer E V, Wagstaff A and Lindelow M. **Analyzing Health Equity Using Household Survey Data, A Guide to Techniques and Their Implementation**

Suggested Readings

- Anthony J. Culyer and Joseph P Newhouse eds., *Handbook of Health Economics,* Elsevier Science, 2000 , Volume 1A & Volume 1B

- Peter Zwefel, Friedrich Breyer and Mathias Kifmann,. Health Economics

Journals

- Journal of Health Economics
- Health Economics
- Health Policy and Planning
- Social Science Medicine: Health Economics
- Bulletin of the World Health Organisation
- The Lancet

Description

- Elective Paper
- Divided into 7 **units**

Teaching Strategy

- Class room Lectures
- Assignment Presentation
- Class Test
- Semester Examination

Aim

To familiarize students with basic concept of health economics and motivate to undertake research/ employment in the field of health economics

What is Health Economics and Why do we Need it?

- Economics studies how scarce resources are allocated among competing uses?

- Health economics applies economic reasoning to health, health care, and health systems.

- The central question: How can limited resources produce the greatest health and social welfare?

Evolution of Health Economics

- Health economics is an interesting paper
- In recent years, there has been increasing attention to health economics research, cutting across the disciplines
- While some universities provides it as elective papers, other introduce as a course

Evolution of Health Economics

- Health economics is a relatively new sub-discipline of economics.
- Growing level of health economics research- health care utilisation, health inequity, health financing, economic evaluation of health care etc

Health Economics in USA

Health Economics is taught globally; largely in school of public health, medical school, economics and health care management

- Harvard University (Elective Paper)

- Johns Hopkins University

- University of California, Berkeley, School of Public Health
- University of Chicago: The Harris School of Public Policy and the Department of Economics

Health Economics in UK

- London School of Economics (LSE) –A specialized Master's in Health Economics, Policy, and Management.

- **University of York** –Centre for Health Economics

- **University of Oxford** – The Health Economics Research Centre (HERC)

- **University College London (UCL)** –Health economics education as part of its public health and economics programs.

Goal

Goal: Better understanding of the **economic aspect** of health and health care

Why to Study Health Economics?

- Gateway to Research
- Health Policy: Health Policy is largely governed by economic consideration and health economics facilitate such understanding
- Multidisciplinary Research :Combine elements of health, economics, statistics, population and public policy
- Cutting-edge research: Increasing global attention
- Better employment prospects: Many organization recruit health economist

Economic Concepts used in Health

- Demand for health services
- Supply of health services
- Price of health services
- Elasticity of health services
- Direct and indirect cost on health care
- Average cost, marginal cost
- Efficiency and equity in health care
- Opportunity cost
- Ability to pay for healthcare
- Out-of-pocket and catastrophic health expenditure on health
- Poverty and impoverishment due to health spending
- Inequity in health and health care utilization
- Gini index and concentration index
- Human capital in health care
- Economic evaluation of health care
- Externalities

Outline

UNIT I: Introduction: SKM

UNIT II: Demand for health and health care: SKM

UNIT III: Health Financing, SKM

UNIT IV: Health Inequity, SKM

UNIT V: Medicare production and cost

UNIT VI: Health Insurance

UNIT VII: Economic evaluation of Health care

Distinguished Health Economist

1. Joseph Newhouse
2. Owen O'Donnel
3. David Cutler
4. Adam Wagstaff
5. Drummond MF
6. Xu K
7. Kenneth J Arrow
8. Allan William

Google Search these authors and describe their work

Add any other leading health economists you came across?

Where do they work?

What are the area they have been working?

Assignment-1

Q1. Write in 500 words on academic contriobution of any of the distinguished health economists

Q2. Does income and consumption data collected in NFHS?

Joseph Newhouse

Incidence of adverse events and negligence in hospitalized patients: results of the Harvard Medical Practice Study I-8186 (1991)
New England journal of medicine 324 (6), 370-376

Health insurance and the demand for medical care: evidence from a randomized experimentWG Manning, JP Newhouse, N Duan, EB Keeler, A Leibowitz
The American economic review, 251-277

Unit I: Introduction to Economics

- What is health and what are measures of health?
- Measures of economic well being?
- Health across world and over time
- Defining health economics
- Relevance of health economics
- Features of economic analyses
- Does Economics Apply to Health and health care?
- Basic concepts in economics
- Is health care different?
- Basic Questions of Health Economics

Unit I: Introduction to Economics

- ❑ Map of health economist
- ❑ Positive and Normative Analyses
- ❑ Efficiency and Equity in health
- ❑ Production Possibility Curve (PPC)
- ❑ Concept of Opportunity Cost
- ❑ Economic Model
- ❑ What is economic gradient of health?
- ❑ Preston Curve
- ❑ Theories of X and Y

Health and Economics

- What is “Health”?
- What are measures of health?
- What is economics?
- What are measures of economic well-being?
- What is the nature of relationship of health and economic variables?

What is ‘Health’?

- **World Health Organisation (1948) :**

  “Health is a state of complete physical, mental and social well-being and not merely the absence of disease or infirmity”

Measures of Health

A. Mortality Measures: Life-expectancy, IMR, U5MR, Adult mortality, premature mortality

B. Morbidity : Hospitalisation rate, out-patient rate, Incidence rate, prevalence rate

C. Subjective Health: How do you rate your health?

D. Activity of Daily Living (ADL/IADL): Perform your daily activities such as bathing, eating, dressing etc

Measures of Health

E. Measures of Nutrition: BMI (underweight, normal and obese)

Malnutrition of children: Underweight, stunting and wasting

F. Summary Measures: DALYs/QALYs

Economics

- Economics is science of scarcity
- Limited resources
- Unlimited “wants”
- It involves “choice”

HEALTH AND ECONOMICS

1. Health is a Priceless commodity

Health being the most precious good.

In order to remain in good health, anything can be done

2. Health care being in crisis

If the cost of health care were to continue to increase at the current rate, than health might become unaffordable to most people.

Kenneth Arrow

Arrow K (1963). Uncertainty and welfare economics of medical care, K Arrow, The American Economic Review, 53(5):941-973

Nobel Laureate- Shared Nobel Prize in 1972

Why is health care economically different?

DISTINCT FEATURES OF HEALTH (Arrow)

1. Uncertainty

2. Asymmetric Information

3.Externalities

Government Intervention is necessary

Conventional Economic Model

Consumer: Demand
Market: Supply
Producer: Production

The consumer has complete knowledge of the product they want to buy
Ex : Phone
Compare prices, specifications, and decide not to buy

Does this model apply to a patient visiting a doctor?

UNCERTAINTY
(Illness, Treatment, and Outcomes)

- Patients do not know if and when they fall sick

- They do not recognize early sign of sickness

Ex: Cancer detection at later stages

- Providers do not know diagnosis with 100% certainty

- They also do not know if treatment works (Outcome)

A Case Study: Example

- Mr X retired from Central Govt service (Group A) two years ago and a diabetic
- He was a fast walker, very active till a week ago; walked over 15 km a day.
- He came from work and felt pain in stomach
- Travelled by local train and reached home
- Start vomiting twice
- Took rest and had dinner
- Went to hospital. ECG test reveal abnormality.
- Referred to Fortis and had surgery with two stent
- Sent back to primary care center and after 2 days got a stroke and paralysed his right face, hand and leg
- Moved to another hospital and life saved

ASSYMETRIC KNOWLEDGE

- Consumer has very little knowledge on disease, medicine and quality of services
- Provider has more knowledge than the consumer
- When we purchase a good service, we have good knowledge about it

EXTERNALITIES

- Externalities is the external benefit or cost

Example

- Pollutant Industries near residence

- Contagious Disease

- COVID 19
- Vaccination

Economics of Health and Health Care

- Health is an interesting economic issue
- There are many questions connected to health which have nothing to do with health care
- How much consumption should society sacrifice to improve health (say increase life expectancy by one year)?
- The weighing of good health against other objectives

Economic Gradient of Health

What are economic variables?

Positive association of health and per capita income

Income growth results in improved health because

- Improvement in Nutrition
- Medical treatment
- Access to clean water and sanitation

Measures of Economic Well Being (Micro)

- **Direct Measures:** Per capita income, Monthly per capita consumption expenditure

- **Indirect measures:** Assets (wealth index), land, house etc. termed as economic proxies

Measures of Economic Well Being

- Per capita income= Household Income/Household Size

- Monthly per capita consumption expenditure (MPCE): Household consumption expenditure/ household size

- Wealth index: Composite variables based on consumer durables, household amenities, land and house

Merits and Limitations of each measures

- **Income:** Underreporting at higher income group and income from multiple sources at lower income
- Less reliable

- **Consumption expenditure:** Less likely to be under reported
- Used to measure economic well being and estimating poverty and inequality in India

Assignment/ HW-2

Q1. Find out the number of variables used in construction of wealth index

Q 2. List out the question on consumption expenditure asked in LASI

Q3. Define improved water and improved sanitation

Wealth Index Now and Than

Assignment

Filmer, D., & Pritchett, L. H. (2001). Estimating Wealth Effects without Expenditure Data-or Tears: An Application to Educational Enrollments in States of India. *Demography*, *38*(1), 115–132. https://doi.org/10.2307/3088292

Extend it with latest data from India. Discuss how the methodology of wealth index has changed since 2001.

Measuring economic well-being from Survey Data

Wealth Index

Identify the economic variables used in explaining health and health care variations from NFHS-5 and NSS health survey.

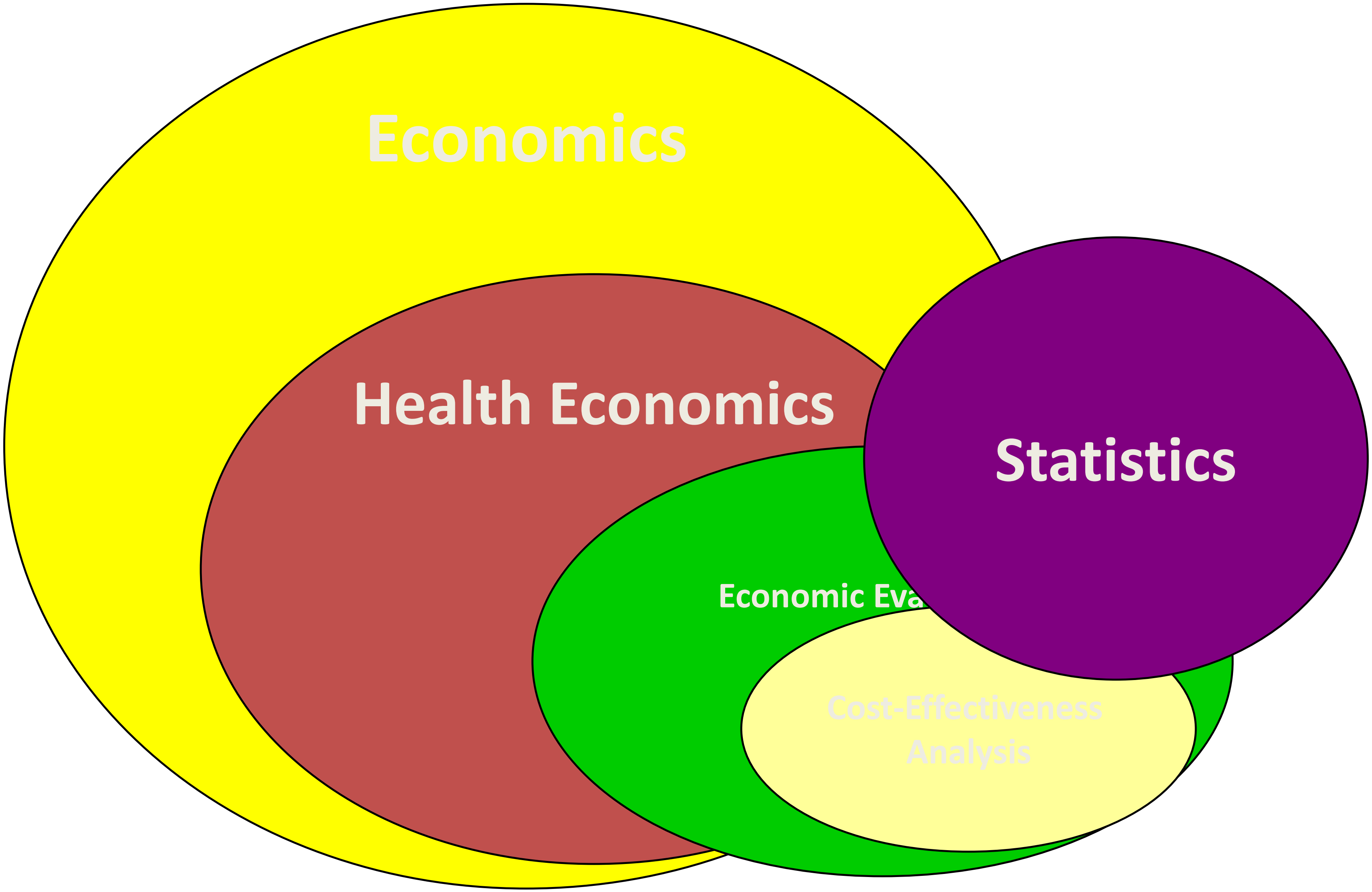
What are the strength and limitations of these variables

Defining Health Economics

- Health Economics is primarily deals with issues related to scarcity of resources in health and health care

Health Economics

- Health economics is an **applied field of economics**
- Drawn from micro economics, public finance, labour and industrial organization
- It is a specialized course with a multidisciplinary approach


Economics
Health Economics
Statistics
Economic Eva
Cost-Effectiveness
Analysis

Health Economics

- Impact of health economics outside the field of economics is immense

- By any criterion, health economics has been a remarkably **successful sub-discipline**

Defining Health Economics

Definition 1:

Health Economics is application of economic theory, models and empirical analyses of decision making by individuals, healthcare providers, and governments with respect to health and health care

Consumer behavior determine health spending

Increase spending on health by Govt can reduce the household spending?

Should COVID-19 testing be done for all?

What is Health Economics

Definition 2:

Health Economics is the **application of microeconomics tools** (utility, demand, cost) to health issues and problems

Definition 3:

"Health Economics studies the **supply and demand of health care resources** and the impact of health care resources on a population" (Medical Encyclopedia)

What is Health Economics?

Definition 4:

Health Economics is the **discipline of economics** applied to the **topics of health** (Moonley G, 1997)

Definition 5:

Health Economics is defined in terms of the **determination and allocation of health care resources**

Health Economics

- How health services should be provided and funded is the key question before researchers, policy makers and political leaders

Three entities

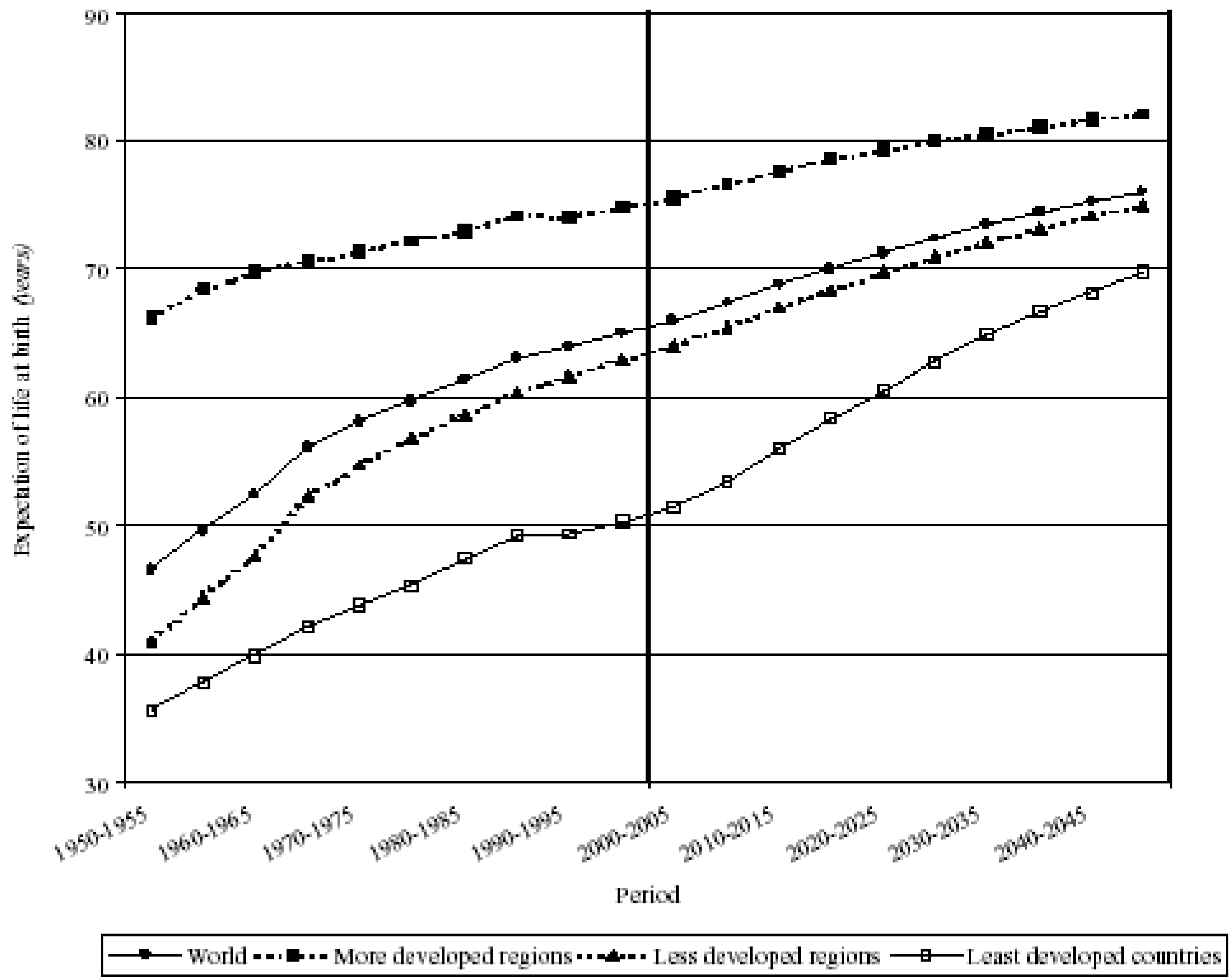
Consumer:	Maximise utility
Business:	Maximise profit
Government:	Maximise Social Welfare

Health Across World and Over Time

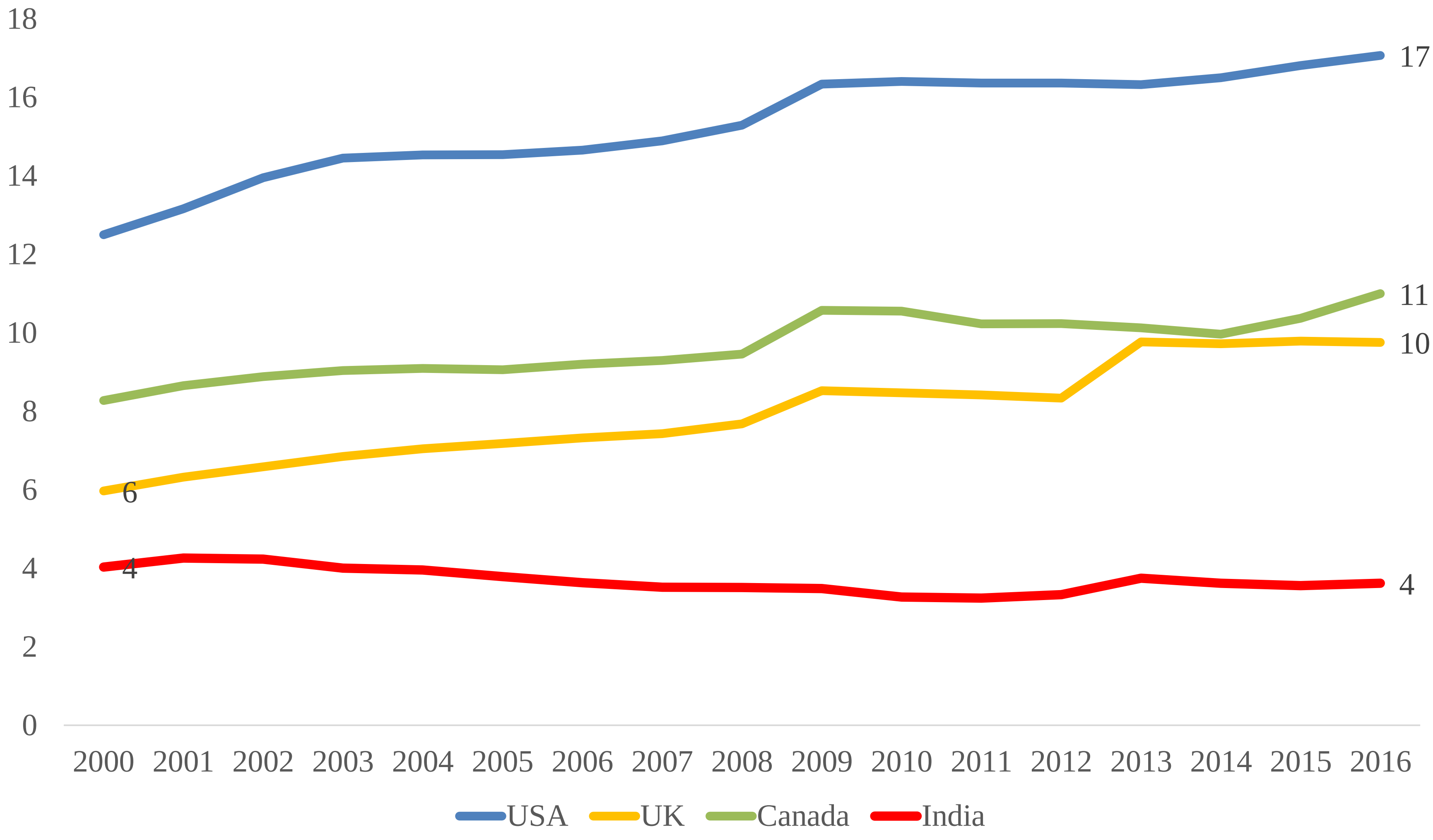
- Health has improved enormously in recent centuries

- At the beginning of 20th century, life expectancy was about 45 years and today it is about 70 years

Life expectancy for the world and the major development groups, 1950-2050



Trends in health expenditure as % of GDP in USA, UK, Canada and India, 2000-16



Source: http://apps.who.int/nha/database/ViewData/Indicators/en

Current Health Expenditure as % of GDP, 2000-16

Sr No	Country	2000	2005	2010	2015	2016	2022
1	United States of America	13	15	16	17	17	17
2	France	10	10	11	12	12	12
3	Germany	10	10	11	11	11	13
4	Japan	7	8	9	11	11	11
5	Maldives	8	9	8	9	11	10
6	Canada	8	9	11	10	11	11
7	Norway	8	8	9	10	11	11
8	Netherlands	7	9	10	10	10	
9	Denmark	8	9	10	10	10	10
10	United Kingdom	6	7	8	10	10	11
12	Finland	7	8	9	10	9	9
13	Greece	7	9	10	8	8	11
14	Iran	5	5	7	8	8	11
15	Hungary	7	8	8	7	7	11
16	Israel	7	7	7	7	7	
17	Nepal	4	5	5	6	6	
18	Colombia	5	6	6	6	6	
19	Viet Nam	5	5	6	6	6	5
20	China	4	4	4	5	5	5
21	Malaysia	3	3	3	4	4	4
22	India	4	4	3	4	4	4
23	Bhutan	4	4	3	4	3	4
24	Indonesia	2	3	3	3	3	4
25	Pakistan	3	3	3	3	3	3
26	Bangladesh	2	2	2	2	2	2

Give interpretation in 5 sentences

Source: http://apps.who.int/nha/database/ViewData/Indicators/en

Income, Health and Health Care across the World

Country	GDP per capita (in $) 2014	Health Expenditures per person(in $)2013	Life expectancy at birth (years) total 2013	Tuberculosis incidence per 100,000 people 2014	Infant mortality rate (deaths per 1,000 live births)2015
Afghanistan	634	55	60	189	66
Albania	4564	240	78	19	13
Bangladesh	1087	32	71	227	31
Brazil	11384	1085	74	44	15
China	7590	367	75	68	9
Ethiopia	574	25	63	207	41
Germany	47822	5006	81	6	3
India	1582	61	68	167	38
Nigeria	3203	115	52	322	69
Norway	64976	6204	81	8	2
Russian Fed.	12376	957	71	84	8
United States	54630	9146	79	3	6

Source: Folland, Goodman and Stano , The Economics of Health and Health Care, P 59

Selected indicators on health spending in USA, 1960-2015

Years	1960	1970	1980	1990	2000	2005	2007	2015
Health expenditure as % of GDP	5.20	7.2	9.1	12.3	13.8	16	16.2	16.8
Nominal per capita health expenditures in $	148	356	1,102	2,813	4,790	6,697	7,421	9,536
Annual rate of growth (%)		10.5	13.00	10.90	5.90	8.90	6.20	3.56
Health expenditures (billion of dollars) in $	$27.50	75	254	714	1,353	1,988	7,421	

Assignment/HW-3

1. Which country has largest per capita health spending?

2. Identify the factors that might lead to rapid growth of health care spending?

3. Compare health spending to food expenditure. How they are similar and different

Economic Analyses

- Economic models are described in words, graph or mathematics

- Economic analyses offers a unique and systematic intellectual framework for analyzing important issues in health care and identifying solution to common problem

Features of Economic Analyses

Four important features are

- Scarcity of societal resources
- Assumption of rational decision making
- Concept of marginal analyses
- Use of economic models

Scarcity of Resources

- Economic analyses is based on the premise that we must give up some of the resources to get some of other.
- Higher allocation of GDP to health care will decrease resources for other use.
- Opportunity cost of health care may be substantial.
- Many may not avail health care as the travel and waiting time is high

Rational Decision Making

- Economic behavior assumes rationality in human behavior
- A rational consumer to use health services
- Investment decision on A Vs B

- Rational decision-making means comparing the incremental costs and incremental benefits of alternative choices and selecting the option that provides the greatest value from limited resources.

Marginal Analyses

- Marginal is used to describe change in benefit or cost that result from one unit increase in production or consumption of a good or services

- It weigh incremental cost against incremental benefits
- It is used in allocation of resources, production, consumer choice, taxation etc

Example

	Prog A	Prog B
Cost per patient	₹1,000	₹2,000
Total cost for 1,000 patients	₹10 lakh	₹20 lakh
Expected QALYs per patient	8.00	8.15
Total QALYs	8,000	8,150

Additional Cost: 10 Lakh
Additional Benefit=150 QALY
Incremental Cost Effectiveness Ratio: 10L/150=Rs 6667

Positive and Normative Analysis

A. Positive Analysis:

It seeks to answer "What is" or "What Happened"

It provides explanation or prediction and free of personal bias

B. Normative Analysis:

Deals with appropriateness or desirability of an economic outcome or policy

It answer questions "What ought to be " or "Which is better"

To reduce out-of-pocket payment, India should adopt a national health insurance program similar to Canada

Give Example

Relevance of Health Economics

- ❑Largest industry
- ❑Economic Constraint
- ❑Public Regulation of Health Sector
- ❑Ethical Significance
- ❑Systematic and Intellectual framework

Largest Industry

- **Largest industry:** Health care is one of the largest industry in developed countries and growing in developing countries.

- Economic performance in health sector is linked to overall economic wellbeing of a country and its citizens

Economic Constraint

Economic Constraint : Decision about how health care is funded, provided and distributed are strongly influenced by the economic environment and economic constraints.

JSY, NHM

COVID-19 and Health Care

Ayusman Bharat

Public Regulation of Health Sector

Public regulation of health sector

National policy concerns resulting from the importance many people attach to economic problems they face in pursing and maintain health

What would be cost of Stent/ PPE kits?

What would be cost for COVID testing?

Ethical Significance

Ethical significance:
Health is extremely emotional issue.

Often there is conflict in looking at economic and ethical way at the allocation of resources both within health care and between health and other goods

Systematic and Intellectual framework

Health care can modify the incidence and impact of ill health and disease.

Economics of health care is a matter of life and death

Did public policy had anything to do in spread of COVID-19 infection/deaths in states of India?

Preston Curve

Preston S (1975). The Changing Relation between mortality and the level of economic development, *Population Studies*, 29(2) 231-248

Demography: Measuring and Modeling Population Processes by Preston et al

Preston Curve: Context

- Relationship of economic condition on mortality has been recognized since ancient times (food supply)
- A movement away from economic determinant of mortality
- Mortality has become increasingly disassociated from economic level because of diffusion of medical and health technologies, facilities and personnel

Preston Curve: Context

To estimate the relative contribution of economic factors to increase in life expectancy during 20th century

Existing Literature

1. Level of Income Influence Level of Mortality at a point of time

Correlation coefficient of GDP per capita and IMR tend to be -0.8

The correlation coefficient of life expectancy and income per head was 0.885 in 1930s and 0.8 in 1960s

2. Level of Income Influence Rate of Change in Mortality

3. Rate of change of Income Influences rate of change of mortality

Measures of Income

GDP Per capita is possibly best economic indicators as it includes all final good and services

Focus of Policy

GDP per capita at constant price is preferred for comparison over time

Preston Curve

1. Objective: To examine the changing relationship of mortality and economic development (cross sectional)

2. Dependent variable: Life expectancy at birth

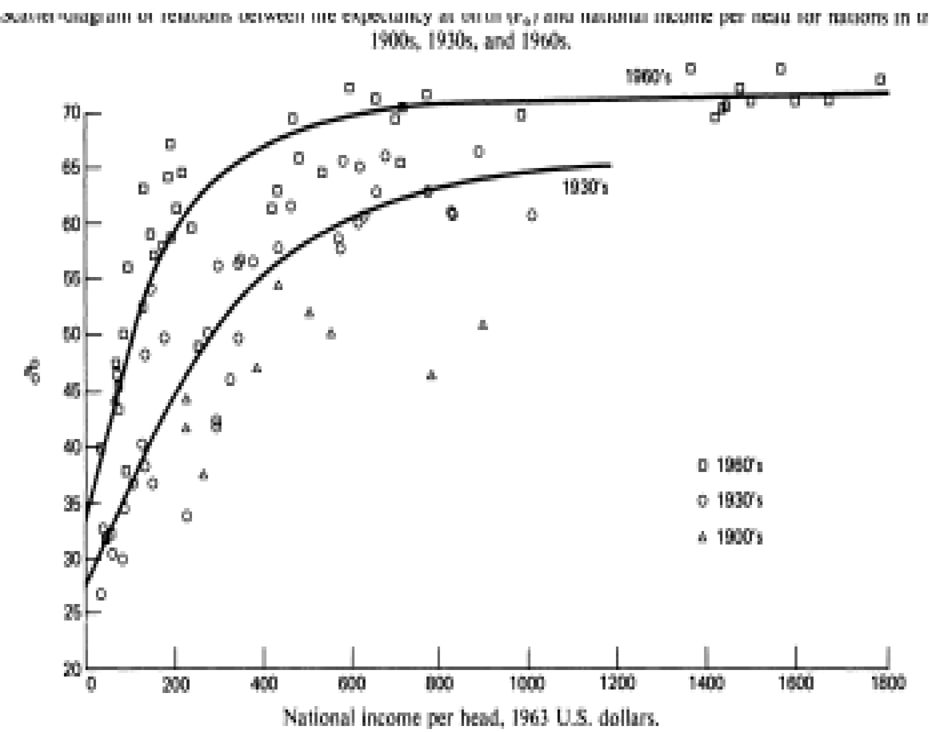
3. Independent Variable: National Income Per head at constant price
4. Relationship: Non-linear

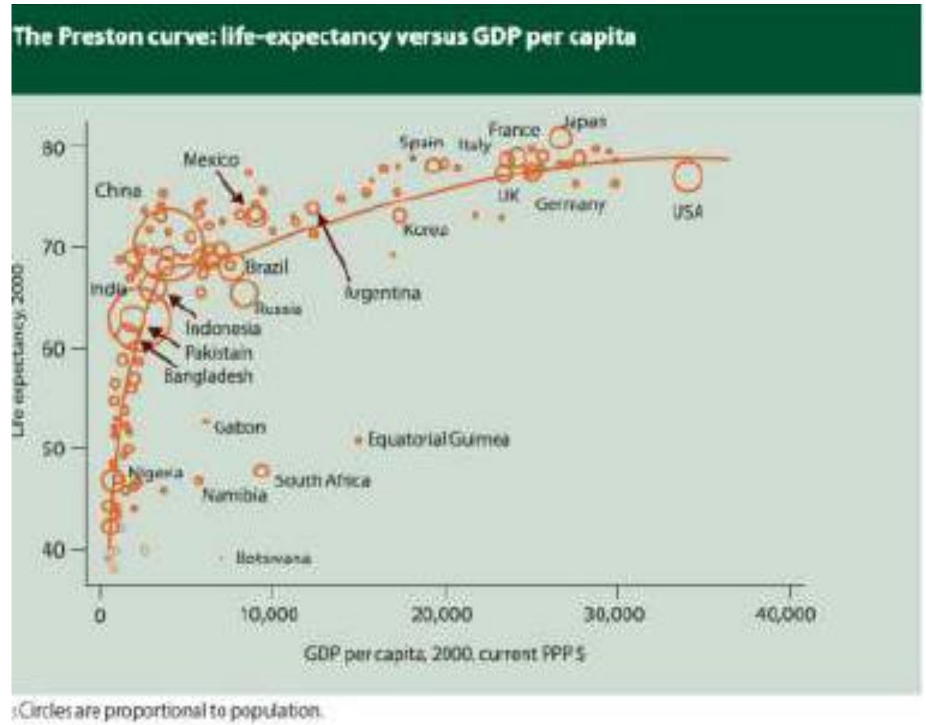
5. Time period: 1900-60

5. Approach: A cross country analyses

6. Method: Scatter plot and logistic curve

Preston Curve





Preston S (1975). The Changing Relation between mortality and the level of economic development, Population Studies, 29(2) 231-248

Life Expectancy

The most common estimation uses a **non-linear regression** or **log-linear model**:

$$\text{LifeExpectancy}_i = \alpha + \beta \ln(\text{Income}_i) + \varepsilon_i$$

- i = country
- $\ln(\text{Income})$ = natural log of GDP per capita
- α = intercept
- β = slope of the curve (effect of log income)
- ε_i = error term

This captures the diminishing returns of income on life expectancy.

LE=a+bln(GDPPCI)+e

Life Expectancy

A common specification looks like:

$$\mathrm{LE}_i \;=\; L_{\max}\,\frac{1}{1+e^{-(\alpha+\beta\ln(\mathrm{Income}_i))}}+\varepsilon_i$$

or equivalently

$$\mathrm{LE}_i = \frac{L_{\max}}{1+\exp[-(\alpha+\beta\ln(\mathrm{Income}_i))]}+\varepsilon_i$$

- LE_i: life expectancy at birth in country i
- Income_i: GDP per capita (often log-transformed)
- $L_{\max}$: asymptotic maximum life expectancy (estimated parameter)
- α, β: parameters controlling the curve's location and steepness
- ε_i: error term

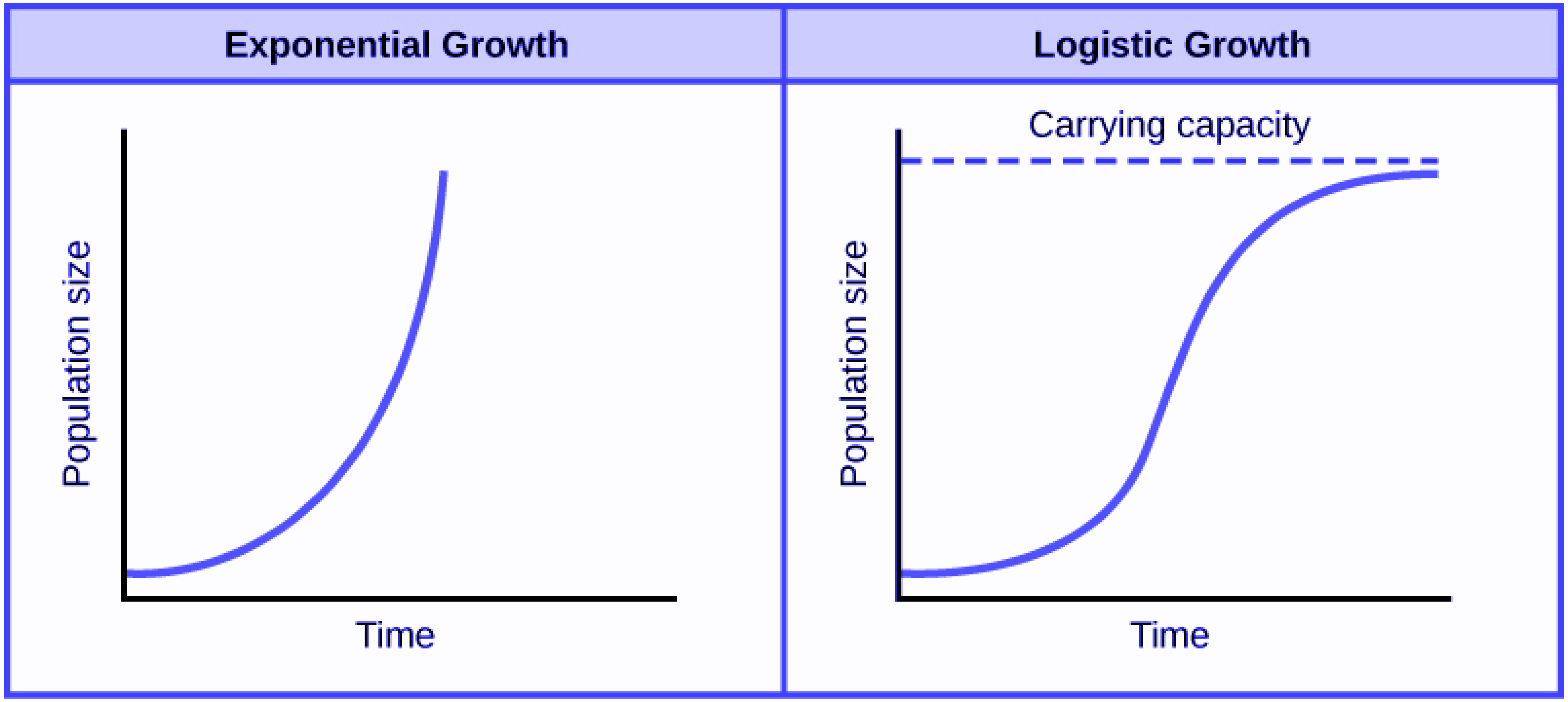
This is essentially a **nonlinear regression**.

Logistic Function

$$f(x) = \frac{L}{1 + e^{-k(x-x_0)}}$$

A logistic curve is a common S –shaped curve (Sigmoid)
Developed as a model of population growth
Initial stage is exponential followed by linear and than stop
f is close to 1 as x tend to +∞
f is close to 0 as x tends to -∞

x_0 = X value of midpoint
L = maximum value
k = growth rate



Fitted Logistic Function of LEB and GDP Per capita

$$1930s: e_0^0 = \frac{80}{1 + exp\{-1.6251 + 2.0768(0.9317)^{Y'}\}}$$

$$1960s: e_0^0 = \frac{80}{1 + exp\{-2.1354 + 2.1697(0.7672)^{Y'}\}}$$

Y'= (Y-37)/ (1012-37) (1930)
Y'= (Y-39.81)/ (2995-39.81)

Where Y is linear transformation to a scale ranging from 0 to 100
The value of 80 in the numerator is specified apriori
Other parameters are derived by least square

Findings

1. The relationship between life expectancy and PCI has shifted upward during 20^{th} century

About 10-12 years of life expectancy at income between$ 100 and $500 and progressively less thereafter

2.Income growth accounts 10-25% in Growth of life expectancy. 75-90% growth in life expectancy between 1930 and 1960 was factors exogenous to income.

$$Z = 21{\cdot}364 - 0{\cdot}01883\ X_1 + 0{\cdot}00128\ X_2 + 0{\cdot}38756\ X_3$$
$$\qquad\qquad (0{\cdot}00990) \qquad (0{\cdot}00588) \qquad (1{\cdot}68985)$$
$$R^2 = 0{\cdot}441$$

where Z = Absolute change in life expectancy, 1930s to 1960s
X_1 = Level of income per head, 1930s
X_2 = Absolute change in income per head, 1930s to 1960s
X_3 = Ratio, income per head, 1930s
(All income figures in 1963 U.S. dollars)

3.Strong positive relationship between national income and levels and life expectancy in poor countries.

In rich countries, the relationship is non-linear and life expectancy is **less sensitive** to variation in average income

4.Mortality has not become progressively dissociated from standards of living at a moment in time

There was no difference between the log-linear correlation coefficients of income per capita and life expectancy

The logistic form actually fits the points in the 1960s somewhat better than it did in the 1930

Assignment

<u>Q.</u> Discuss the changing relationship of per capita income and life expectancy of birth across countries and over time (1980-2021). Discuss the relevance of Preston Curve.

Hint: Use data from World Bank/WHO. Use per capita income at constant prices. Prepare a stata file. Interpret result

Ref: Preston S (1975). The Changing Relation between mortality and the level of economic development, Population Studies, 29(2) 231-248.

Data Base-Global

http://apps.who.int/nha/database/Select/Indicators/en

https://databank.worldbank.org/reports.aspx?source=2&series=SP.DYN.LE00.MA.IN&country=#

http://www.worldbank.org/en/topic/health/publication/analyzing-health-equity-using-household-survey-data

CONCEPTS: EFFICIENCY and EQUITY

Efficiency and **equity** are at the heart of the Health Economics

Efficiency of services, goods or activity

Efficiency is optimal production and allocation of resources given existing factors of production

CONCEPTS: EFFICIENCY and EQUITY

Three type of efficiency:

1. Technical Efficiency

2. Cost-effective efficiency

3. Allocative efficiency

Supply-side Efficiency

Demand Side efficiency

CONCEPTS: EFFICIENCY

Technical efficiency : **Minimise inputs** to produce a **given level of output**

Cost effective efficiency: **Minimise the cost** of producing a **given level of output**

Technical Efficiency: Production Function

Input	Process 1	Process 2	Process 3	Process 4
L	2	3	1	1
K	3	2	4	1

Which process in efficient in producing same level of output?

Four Basic Questions

1. What combination of **medical and non-medical goods** and services to be produced in the **economy?: Allocative efficiency**

2. What particular **medical goods and services** should be produced in the **health economy**?:**Allocative efficiency**

3. What specific health care **resources should be used** to produce the final goods and services? : **Production/ Technical efficiency**

4. **Who should receive** the medical goods and services?: **Equity**

Why Are We Concerned?

- Efficiency questions in health care sector arise because costs are high.

- Equity questions arise because many people are uninsured or under insured.

EFFICIENCY in Healthcare

- Decision makers are increasingly faced with the challenges of reconciling growing demand for health care with limited funds

- Economist argued that achieving greater efficiency from scare resources should be a major criteria for priority setting

- Efficiency measures how successfully the inputs have been transformed to output (crude measure: output/input)

- Efficiency means whether healthcare resources are being used to get the best value of money

- It is the optimal use of resources to achieve a specific outcome

CONCEPTS: Allocative EFFICIENCY

- **Allocative efficiency** incorporate demand side or consumption factor.
- Allocative efficiency is achieved when resources are produced and allocated so as to produce the optimal level of each output and **distribute** the output in line with the value consumer place on them
- The Problem: How to distribute resources with a focus on health care
- Allocative efficiency is achieved when resources are allocated to maximise the welfare of the community

Definition Pareto Efficiency

An economically efficient (optimal) outcome in society is one under which it is impossible to make someone better off without making someone worse off.

Scarcity and the Production Possibilities Curve (PPC)

To get something, one must usually give up something else

Presented by PPC/ PPF

A PPC shows the trade-off between any two goods given a fixed stock of resources and **technology**

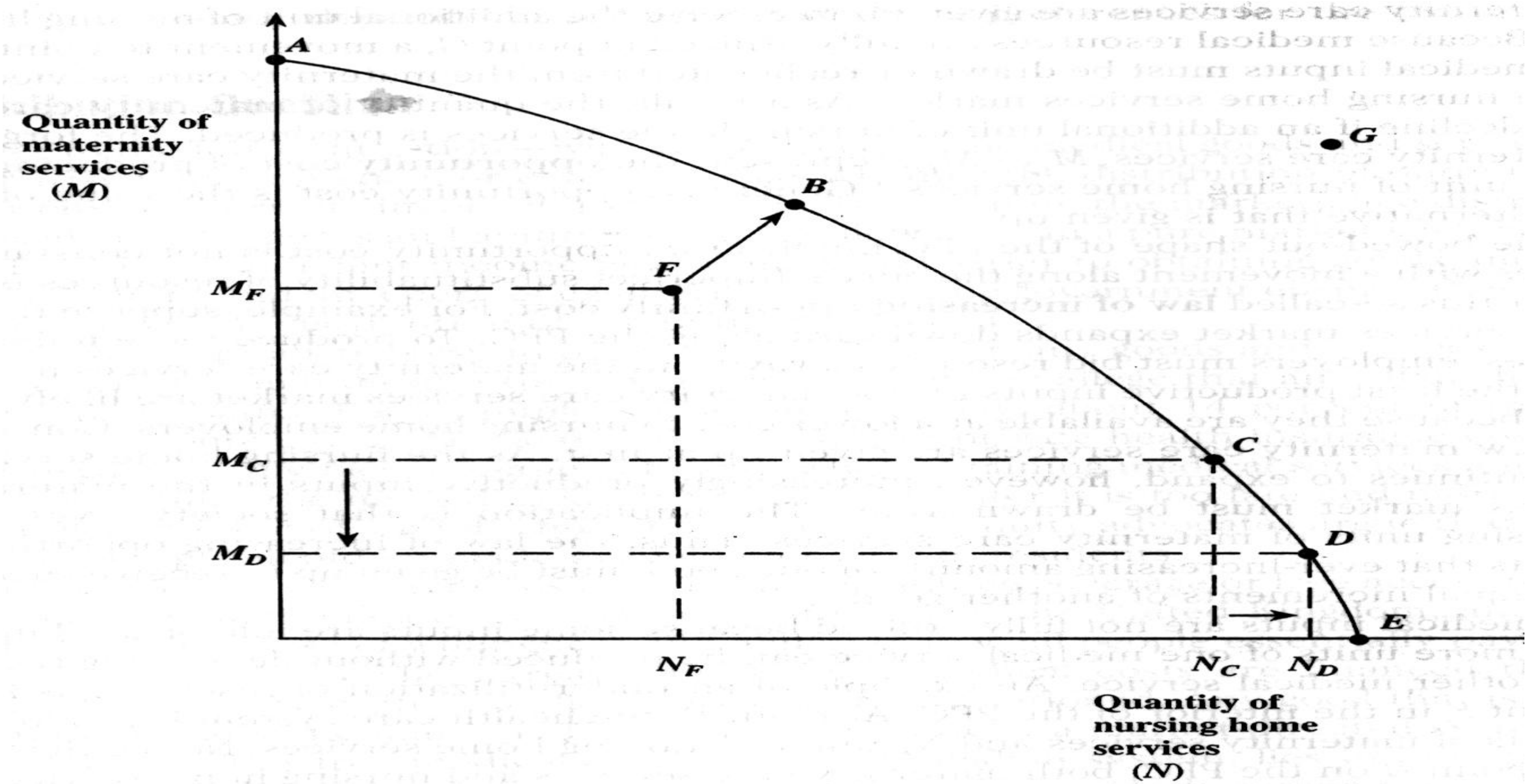
We can not have all of everything we want
We face resources constraint and must trade-off is central to PPF

Production Possibilities Curve (PPC)

A PPC is an economic model that depicts the various combinations of any two goods and services that can be produced efficiently given the stock of resources , technology and various institutional arrangements.

1. PPC is hypothetical
2. Shows opportunity cost of resources
3. Efficiency of resources

Production Possibility Curve for Maternity and Nursing Home Services



Point F is inefficient
Point G is infeasible

Production and Allocative Efficiency: Production Possibilities Curve (PPC)

Production efficiency is attained when the health economy operates **at any point on the PPC**

Allocative efficiency is attained when society chooses the **best or most preferred point** on PPC

All point on the PPC are possible for allocative efficiency but **the ideal or optimal point** depends on society's underlying preferences for the two medical services

Concept : Opportunity cost

Second best alternative

"Opportunity cost is the benefit forgone which could be obtained from a resource in its next-best alternative use."

Example of opportunity cost

Paediatric Care (No Children Treated in ‘000’s)	Care of Elderly (No of Elderly Treated in ‘000’s)	Opportunity Cost of Treating Children in Terms of Elderly Patients Forgone
0	30	0
1	28	2
2	24	6
3	18	12
4	10	20
5	0	30

Implications of opportunity cost

- Deciding to ***do*** A implies deciding ***not*** to do B (i.e. value of benefits from A>B).
- Cost can be incurred without financial expenditure.
- Value not necessarily determined by “the market”.

Society's Trade-off Between Maternity and Nursing Home Services

Point	Quantity of maternity services	Quantity of nursing home services	Opportunity cost of nursing home to maternal services
A	870	0	
B	795	100	75
C	654	200	141
D	501	300	153
E	321	400	180

THEORIES OF X AND Y OF HEALTH ECONOMCIS

Discussion concerning health economics spark considerable debate

Musgrave (1995) provided a useful conceptual framework in which different views can be contrasted; known as **the theories X and Y of health economics**

THEORIES OF X AND Y OF HEALTH ECONOMCIS

VIEW	Theory X	Theory Y
Determinant of Health	Health and Disease Occur randomly	Health is determined by people’s lifestyle choices
Medical care	Special Necessity, inelastic nature of demand, consumer ignorance make medical care unique	No different from any other goods and services

THEORIES OF X AND Y OF HEALTH ECONOMCIS

VIEW	Theory X	Theory Y
The practice of Medicine	A Science (Expert will arrive at the best way to treat each and every illness)	An art (Many illness are patient specific. Provider will never find the best cure for a given illness)
Financial Rewards	Financial rewards diminishes the quality of caring.	Financial rewards are responsible for generating high quality medicine

THEORIES OF X AND Y OF HEALTH ECONOMCIS

VIEW	Theory X	Theory Y
Policy	Financial rewards are the source of system failure. Regulation are needed to mitigate economic forces	Reduce regulations and encourage market forces Healthcare markets are overregulated.
	Tax the healthy, subsidies the risk	Tax the sick, not the healthy
	Discourage new medical technologies	Encourage new medical technologies

Use of regression and economic models

An economic model describes a hypothesized relationship between **<u>two or more variables</u>**

Ex: Health care expenditure (E) are directly related to income (Y)

$E= f\ (Y)$

Health care expenditure are expected to rise with income

Health Expenditure

Assumptions: All factors other than the variable of interest remains unchanged

Mathematical Form of expenditure function is

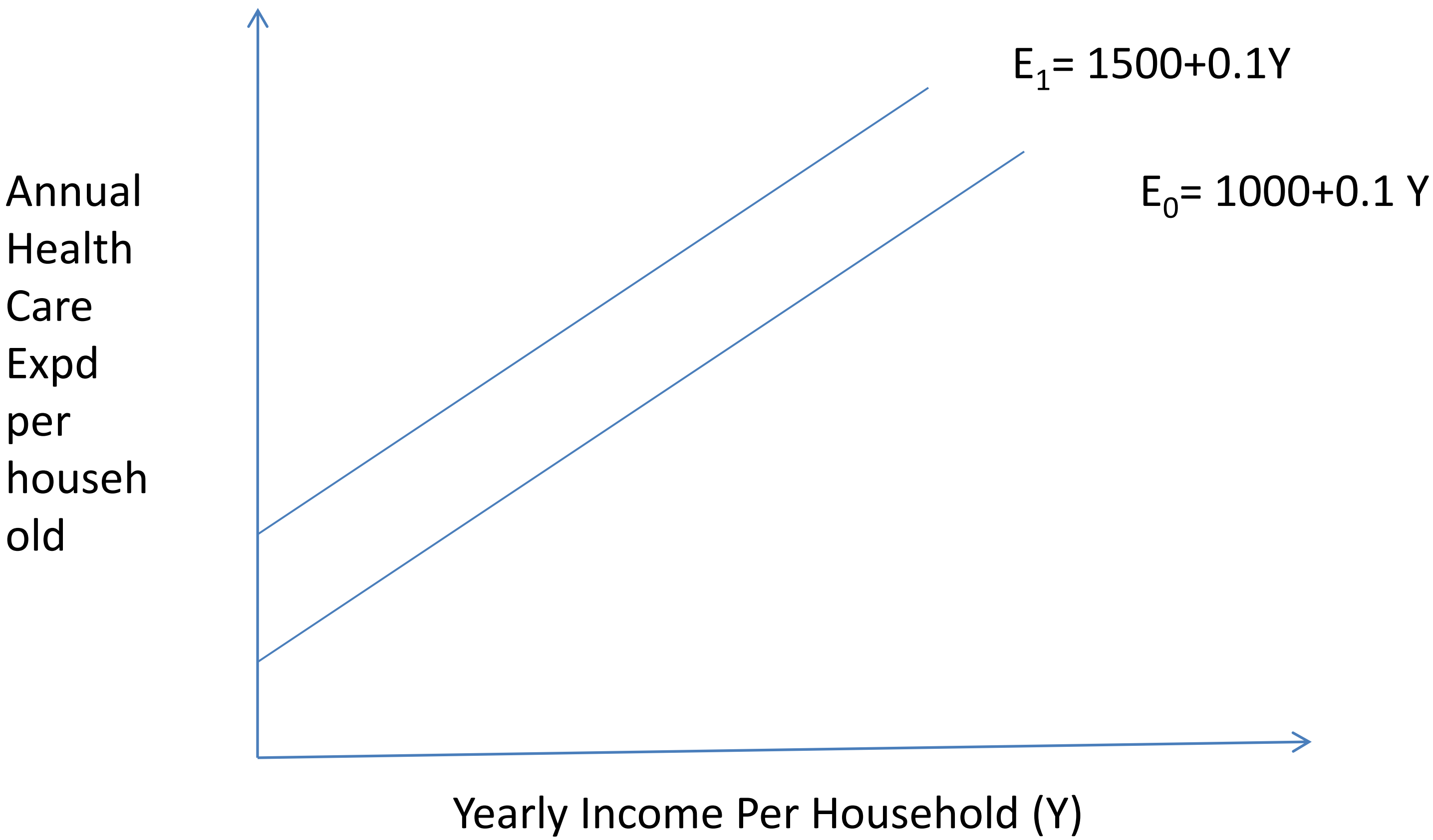
$E = a + bY$ where a and b are constant

HE related to income in a linear fashion

$E = 10 + 0.1Y$

When $Y=0$, $E=10$.

Shift in Health Care Expenditure



Increase in health care expenditure due to factors other than income

An empirical study provides useful information on how strongly income affects health care spending

Review Questions

Q 1. Define is Health Economics? Explain the road map of Health Economics. Discuss the relevance of the subject in context of population studies.

Q 2. Define efficiency and equity. How efficiency and equity are related? .Explain equity and efficiency using PPC.

Q3. Define is Health Economics? Explain the road map of Health Economics. Discuss productive and allocate efficiency using the production possibility curve (PPC) in context of Maternity Care and Nursing home.

Q 4. Discuss the opportunity cost. Discuss the opportunity cost of pediatric and geriatric care

Q 5. Define economic model. Discuss and interpret the expenditure function $E= 1000+0.07\ Y$ where E is the health expenditure and Y is the income.

Review Questions

Q6. Discuss the basic questions confronted by health economist. Discuss the positive and normative analyses with example. Discuss and explain Musgrave's Dichotomy on theories of X and Y in five dimension of health with example.

Q7. Discuss direct economic measures used in health. Explain two way causation of health and income. Discuss the salient findings of Preston curve and its relevance

Q8. Is health care different? Discuss the distinctive features of the health economy. Examine each one separately and describe other commodities that share those features.

Homework 5

Define production possibility curve. Draw a production possibility curve between health and all other goods. Insert a point in the drawing that illustrate an economy with an inefficient health system. Insert two additional points that illustrates two-efficient economies but two that contrast in their relative emphasis on health care versus all other goods. Is there a cost to society of policies that lead to increase in health care? Explain.